

Customer Shopping Behavior Analysis

Insightful Analytics Report • Python + PostgreSQL + SQL + Power BI

Executive perspective

This project is an end-to-end customer analytics workflow designed to turn a shopping-behavior dataset into decision-oriented analysis. It combines data preparation in Python, relational analysis in PostgreSQL, business-question-driven SQL, and a Power BI dashboard layer.

Important interpretation note: this report only states numeric findings that are directly preserved in the supplied project files. The SQL file defines ten business questions, but its result tables are not embedded in the supplied SQL. Accordingly, this report does not invent query results.

Verified dataset snapshot

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Dataset records	3,900
Original columns	18
Age range	18–70
Average age	44.07 years
Average purchase	\$59.76
Median purchase	\$60
Average previous purchases	25.35
Average review rating	3.75 / 5
Review ratings missing before cleaning	37
Product categories	4

1. What the data says

The notebook documents a dataset of 3,900 customer records with 18 original fields. The variables cover demographics, products and categories, purchase amount, location, size, color, season, review rating, subscription status, shipping, discount and promotion usage, previous purchases, payment method, and purchase frequency.

The typical transaction is close to \$60: the mean purchase amount is \$59.76 and the median is \$60. The standard deviation is \$23.69, while observed purchases range from \$20 to \$100. This shows meaningful variation in transaction value around a central level near \$60.

Customer age averages 44.07 years, with observations spanning ages 18 to 70. Previous purchases average 25.35, with a range from 1 to 50. These variables support lifecycle segmentation, although the project uses previous purchases rather than a complete longitudinal customer history.

Review ratings average 3.75 out of 5. Before cleaning, 37 of 3,900 review-rating records were missing. The notebook fills missing ratings using the median rating within each product category, resulting in zero missing values after cleaning.

2. Data preparation and analytical design

The preprocessing stage makes the source data consistent and analysis-ready. Column names are converted to lowercase with underscores, and the purchase amount field is standardized to `purchase_amount`.

Age is divided into four quartile-based groups: Young Adult, Adult, Middle-aged, and Senior. Purchase-frequency labels are also translated into approximate day intervals such as 7 days for Weekly, 14 for Fortnightly/Bi-Weekly, 30 for Monthly, and 90 for Quarterly/Every 3 Months.

The transformed DataFrame is designed to feed PostgreSQL, where the SQL layer performs grouped aggregation, conditional logic, subqueries, CTEs, and window functions.

3. Business questions covered

Analysis area	Business question
Revenue mix	Compare total revenue generated by male and female customers.
High-value discounted buyers	Identify discounted purchases at or above the overall average purchase amount.
Product quality	Find the five products with the highest average review rating.
Shipping economics	Compare average purchase amount for Standard versus Express shipping.
Subscription value	Compare customer count, average spend, and total revenue for subscribers and non-subscribers.
Discount intensity	Identify products with the highest percentage of purchases receiving discounts.
Lifecycle segmentation	Count New, Returning, and Loyal customers based on previous purchases.
Category winners	Find the three most purchased products within each category.
Repeat-buyer subscription	Examine subscription status among customers with more than five previous purchases.
Age-group revenue	Compare total revenue contribution across age groups.

4. Insight opportunities

- **Customer value:** Purchase amount and previous purchases support a two-dimensional view of value: transaction size versus demonstrated repeat behavior.
- **Promotion effectiveness:** The discounted-purchase analysis can identify high-value purchases made with discounts, but this association cannot establish that discounts caused higher spending.
- **Subscription behavior:** Comparing subscribers and non-subscribers on both average spend and total revenue separates transaction value from customer-scale effects.
- **Product portfolio:** Using both purchase volume and review rating distinguishes products that sell frequently from products that receive stronger customer ratings.

- **Customer lifecycle:** The New/Returning/Loyal framework converts previous-purchase counts into reproducible customer segments.
- **Demographic contribution:** Age-group revenue analysis can show where revenue is concentrated, while quartile-based groups keep segment sizes comparatively balanced.

5. Analytical strengths

The project has a clear progression from raw data to business questions. The category-level median imputation is targeted rather than global; engineered age and purchase-frequency features support segmentation; and the SQL layer uses practical analytical patterns such as CTEs, conditional aggregation, subqueries, and ROW_NUMBER() with PARTITION BY.

The SQL questions also go beyond simple aggregation. They examine high-value discounted customers, rank products within categories, compare subscription economics, and connect repeat purchasing with subscription status.

6. Limitations and interpretation cautions

- The supplied SQL file contains query definitions but not their result tables, so specific query-result values are not claimed here.
- The dataset is structured around customer shopping records rather than a documented longitudinal transaction history.
- The purchase-frequency-to-days mapping is an analytical convenience based on categorical labels.
- The discounted-versus-average analysis describes observed association and should not be interpreted as causal evidence.

7. Recommended next analytical steps

1. Execute the SQL against the prepared PostgreSQL table and capture the result tables in the project documentation.
2. Add revenue-share measures for gender, subscription status, and age group.
3. Compare discounted versus non-discounted spending, ideally controlling for product category.
4. Combine purchase volume, average rating, and discount rate into a product-performance matrix.
5. If transaction dates become available, extend the model into cohort, retention, seasonality, and customer-lifetime-value analysis.

8. Executive conclusion

The project establishes a practical foundation for understanding customer shopping behavior. The verified dataset contains 3,900 records, a \$59.76 mean purchase amount, a \$60 median purchase, an average customer age of 44.07, an average of 25.35 previous purchases, and an average review rating of 3.75.

The main strength of the project is the connection between technical data work and concrete business questions. The next step is to capture the SQL and Power BI outputs so the project can move from an analytical framework to a fully quantified business story.

Source basis

Prepared from the supplied Customer_Shopping_Behavior_Analysis.ipynb and Customer_Shopping_Behavior_Analysis.sql project files. No external tutorial or creator reference is included.